Project Requirements Document

BI Analyst: Hasna AIT BOUI

Client/Sponsor: Retail chain Company

The main purpose of this project is to enable stakeholders to monitor sales trends and profitability in order to make reliable, data-driven decisions regarding product and category strategies. In addition, teams need actionable insights into customer behaviors, segmentation, and demographics to build targeted marketing campaigns. Finally, the tool will allow management to track store performance and identify underperforming locations for targeted operational improvement. Investing resources in this BI solution will help the company optimize discount spending, increase customer retention, and drive overall profitability across all stores.

Key Dependencies:

- Project Team & Primary Contacts:
    - BI Team: Hasna AIT BOUI (Lead BI Analyst) responsible for data modeling, ETL, and dashboard development.
    - Executive Sponsors & Stakeholders: Chief Revenue Officer (CRO), Head of Marketing, Regional Store Managers, Product Category Managers, and the BI Director for business requirements, validation, and sign-off.
- Technical & Data Dependencies:
    - Access to raw operational data tables: Customers, Products, Stores, and Transactions.
    - Google BigQuery environment configured for data transformation and star schema modeling.
    - BI visualization tool (Looker Studio, Power BI, or Tableau) connected to BigQuery.
- **Expected Deliverables:**
    - Business documentation: Stakeholder Requirements Document (SRD), Project

Requirements Document (PRD), and BI Strategy Document.

- ETL pipeline and star schema in BigQuery.
- An interactive BI dashboard tracking core sales, customer, store, and product KPIs.

Stakeholder requirements:

In order to increase profitability across all stores, the dashboard must help the company understand customer behavior and provide up-to-date visibility into sales trends and store performance to intervene immediately if sales drop or stores underperform.

The dashboard requirements are prioritized as follows:

- **Required:** Scorecards or summary cards showing updated Total Revenue, Total Profit, and Average Order Value (AOV).
- **Required:** Interactive global filters for Date Range, Store Location, Product Category, and Demographics.
- **Required:** A chart showing sales and profit trends over time (daily, monthly, quarterly).
- **Required:** A chart or table displaying top and bottom products and categories by revenue and profitability.
- **Required:** A comparative view evaluating store and regional performance.
- **Required:** An analysis of promotional discount impact on overall sales and profit margins.
- **Required:** Charts that show sales by customer age, gender, and city to understand their influence on purchasing habits.
- **Desired:** A chart showing the most used payment methods.

The BI tool will be an interactive dashboard that allows all stakeholders to track all required sales, store, and customer metrics. As a result, they will be quickly aware of any underperformance in sales, profitability, or store operations so they can intervene promptly. The dashboard will also help them gain insights into customer behavior through segmentation and demographics, allowing them to build targeted marketing strategies.

Success will be measured using the following SMART criteria:

- **Specific:** Build an interactive dashboard backed by a BigQuery star schema that covers sales, store performance, product categories, and customer demographics.
- **Measurable:**
  - 100% data accuracy between the source tables and the final dashboard.
  - Dashboard load and filter interaction response time under 5 seconds.

    - Daily automated ETL refresh running without errors.
- **Achievable:** The four source tables (Customers, Products, Stores, Transactions) are already available and ready to model.
- **Relevant:** Directly enables leadership to track discount impacts, compare store performance, and target profitable customer groups.
- **Time-bound:** Deliver documentation week 1, ETL modeling week 2 and 3, and dashboard week 4.

Current User Experience: Currently, users only have access to raw data across four separate tables: Customers, Products, Stores, and Transactions. Because the data is not connected or visualized, stakeholders cannot quickly gain meaningful insights or evaluate specific performance metrics without manual effort and delays.

Ideal Future Experience: With the new interactive dashboard, stakeholders will be able to easily see which stores perform well, track overall sales and profitability, and identify which products and categories generate the most profit. They will also have immediate access to insights on customer behavior whenever they need to build and adjust targeted marketing strategies.

Data Quality & Integrity: We assume that the Customers, Products, and Stores tables have minimal missing values and reliable primary keys, and that any duplicates or formatting inconsistencies will be resolved during the initial ETL phase.
Transactional Data: We assume the Transactions table records valid transaction IDs, accurate timestamps, non-negative quantities, and realistic discount ranges.
System & Environment Access: We assume that Google BigQuery and the BI visualization tool will be accessible throughout the project to build the star schema and automate data updates.
Stakeholder Feedback & Alignment: We assume stakeholders will be available to review and validate a dashboard mockup and confirm metric definitions before final deployment.
Customer Privacy: After cleaning the Customers table, the reporting layer will expose only necessary analytical attributes (Customer ID, Gender, Date of Birth / Age, City, and Join Date). Direct personal identifiers (such as full names, email addresses, or phone numbers) will be excluded to protect customer privacy and comply with data privacy standards.

Role-Based Access Control (RBAC): Access to dashboards and underlying data will be restricted based on stakeholder roles and responsibilities:

- **Executive Leadership (CRO):** Full access to high-level financial metrics, net profitability, and discount impact across all stores and regions.
- **Marketing Team:** Access to aggregated customer demographics, segmentation, and purchasing behavior views.

- **Regional Store Managers:** Filtered access restricted to store performance and localized operational comparisons.

Accessibility: (List key considerations for creating accessible reports for all users.)

The dashboard will be built following a Z-pattern layout, placing the most important summary metrics and drop-down filters (Date, Store, Product Category, and Demographics) at the top for easy navigation.

The dashboard will use high-contrast colors, legible font sizes, and clear chart labels to ensure strong readability.

The project will be delivered in four structured phases over a four-week timeline:

- **Week 1 (Part 1 - Alignment & Documentation):** Meet with stakeholders to gather business needs, then draft and finalize the Stakeholder Requirements Document (SRD), Project Requirements Document (PRD), and BI Strategy Document for stakeholder validation.
- **Week 2 (Part 2 - Data Preparation & ETL Architecture):** Ingest raw data tables (Customers, Products, Stores, Transactions) into BigQuery, perform data cleaning, and design the star schema.
- **Week 3 (Part 2 - ETL Pipeline & Metric Calculation):** Build and test the automated daily ETL pipeline in BigQuery, implementing all calculated fields and core business KPIs.
- **Week 4 (Part 3 - Dashboard Development & Delivery):** Build the interactive dashboard in Looker Studio/Power BI/Tableau with global filters, validate metrics with stakeholders, and complete the final handover.